

Patrick O'Brien

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PROFESSIONAL SUMMARY

Software Engineer with a B.S. in Computer Science (2025) and hands-on experience building full-stack web applications and production automation tools. Comfortable across the stack: React front ends, FastAPI/Flask APIs, SQL data layers, and Python automation against cloud APIs, with CI/CD pipelines and automated test coverage throughout.

SKILLS

Languages: Python, TypeScript, JavaScript, PowerShell, SQL, Java, C++, C, HTML/CSS

Frameworks: React, FastAPI, Flask, SQLAlchemy

Tools & Technologies: Git/GitHub, GitHub Actions (CI/CD), REST APIs, JWT Auth, pytest, Microsoft Graph API, Microsoft 365, Microsoft Entra ID, Azure

Databases: PostgreSQL, MySQL, SQLite, Microsoft Access

Core Concepts: Object-Oriented Programming, Data Structures, Web Development, Automated Testing

Operating Systems: Linux, Windows, macOS

EXPERIENCE

IT Support Specialist, Software Automation Focus

Jun 2025 – Present

Habitat America

Annapolis, MD

- Building a Python automation tool using the Microsoft Graph API to fully automate user provisioning and deprovisioning across 100+ properties and 400+ users, eliminating manual account setup and reducing onboarding to a single automated workflow.
- Administer and maintain Microsoft 365, Microsoft Entra ID, and Exchange environments supporting 400+ users across 100+ managed properties.
- Diagnose and resolve technical issues across systems and applications, applying structured, engineering-driven problem-solving under time constraints to keep operations running smoothly.

Database Administrator

Mar 2019 – May 2025

Mike's Mowing (family business)

Ridgely, MD

- Independently designed, built, and maintained a Microsoft Access database for a family-owned landscaping business, covering customer records, service history, and scheduling.
- Owned data integrity across the system: validating inputs, resolving inconsistencies, and structuring tables, queries, and forms to keep records accurate and reliably accessible over a 6-year period, while working full-time alongside coursework.

Job Coach

Mar 2013 – Mar 2019

Benedictine

Ridgely, MD

- Provided on-the-job coaching and support for individuals with disabilities in workplace settings.

PROJECTS

Ticket Management System github.com/pobriendev/ticket-system

React (Vite), FastAPI, PostgreSQL, JWT, GitHub Actions

- Architected a full-stack helpdesk ticketing application with JWT authentication, role-based access control, and an automatic audit trail on every status and assignment change.
- Designed the data model and REST API (FastAPI + SQLAlchemy/Alembic) covering ticket CRUD, filtering, pagination, comments, and category management.
- Wrote 171 backend tests (~96% coverage) and 128 frontend tests with pytest and Vitest, and built a CI/CD pipeline (GitHub Actions) running tests and lint on every push, deploying on merge to main.

Employee Provisioning Tool github.com/pobriendev/employee-provisioning-tool

Python, Microsoft Graph API, YAML

- Built a Python CLI that automates Entra ID employee onboarding and offboarding end-to-end: account creation or role-account reuse, license assignment, group membership, and credential delivery, replacing a manual multi-step portal process.
- Designed app-only (client-credentials) authentication scoped to least-privilege Graph permissions, with dry-run mode, lockout-first offboarding, and a full audit log for every write action.

Entra Stale Accounts github.com/pobriendev/entra-stale-accounts

Python, Microsoft Graph API, PyPI

- Published a read-only CLI to PyPI (pip install entra-stale-accounts) that flags inactive Microsoft Entra ID accounts past a configurable threshold, including never-signed-in accounts.
- Generalized the tool from production account-lifecycle automation into a standalone package with a mocked test suite, CI via GitHub Actions, and automatic Graph API retry/backoff.

Ski Resort Explorer github.com/pobriendev/ski-resort-explorer

React, Flask, MySQL, OpenWeatherMap API

- Architected and built a full-stack web application from scratch in React and Flask, exposing RESTful endpoints for search, filtering, and data aggregation, with a MySQL/phpMyAdmin schema for fast search.
- Integrated a third-party API (OpenWeatherMap) to aggregate and serve real-time weather conditions and precipitation radar data.

EDUCATION

Delaware State University

B.S. in Computer Science. GPA: 3.57.

2025

Delaware Technical Community College

A.S. in Computing and Information Science.

2020